

The Tech Stack & Cost Optimization Brief

One-page architecture defense for audit-grade MRV across fragmented Indian farms while keeping platform spend under **\$10 / Rs 830** per farm.

RECOMMENDED STACK

Edge Layer

React Native keeps the product Android-first for low-end devices with direct GPS, camera, and background worker access.

WatermelonDB + SQLite is the required offline layer: it stores polygons, evidence manifests, and sync checkpoints on-device until a stable 4G window appears.

Core Backend

Supabase + PostgreSQL/PostGIS remains the source of truth for auth, storage, spatial records, and dashboard read models.

Node.js / Edge Functions handle idempotent ingestion, event projection, evidence validation, and payout-trigger workflows.

Spatial + ML

H3 indexing makes overlap detection and national heatmaps cheap even with millions of micro-landholdings.

Sentinel-2 + ISRO Bhuvan APIs provide low-cost satellite context for tree-growth review and field-edge anomaly checks.

WHY WATERMELONDB

- Supabase is strong as a hosted backend, but it does not natively solve rural offline sync or replay-safe polygon uploads.
- WatermelonDB acts as the **edge database** and keeps an append-only local event log in SQLite.
- A background sync worker uploads only when connectivity, battery, and GPS quality are acceptable.
- Each event carries an **idempotency key**, so retries cannot create duplicate polygons or duplicate biochar rows.

CHEAP ACCURACY

- Use guided perimeter walks with multi-sample GPS averaging, corner pauses, and outlier rejection.
- Store both the raw trace and simplified polygon so auditors can replay how a boundary was captured.
- Run server-side QA against Sentinel-2 imagery and Bhuvan layers to flag implausible edges or area changes.
- Use H3 overlap scans before approval to block double counting across adjacent or duplicated claims.

WORKFLOW & COST DEFENSE

STAGE	DESIGN CHOICE	WHY IT STAYS CHEAP
Farm mapping	React Native + WatermelonDB + GPS averaging	No survey-grade hardware; only commodity Android handsets and offline storage.
Spatial storage	PostGIS + H3	Open-source geospatial stack instead of proprietary land survey APIs.
Evidence capture	Live camera only + on-device EXIF hashing	Prevents gallery spoofing without buying managed hardware.
Remote QA	Sentinel-2 + Bhuvan + rules engine	Open or low-cost data supports canopy and land-change checks at scale.
Cloud	AWS or GCP India region	Usage-based compute with DPDP-aligned data residency.

< \$10
Target platform cost ceiling per farm.

\$0
Additional survey hardware required per farm.

Open Data
Sentinel-2 and Bhuvan reduce recurring imagery spend.

Zero-Trust
Live capture, geofence checks, and chained events protect credit quality.

Decision summary: keep Supabase as the system of record, but add **WatermelonDB** on-device because Joita's hardest problem is edge sync, not cloud storage.